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28 February 1977

MILITARY SPECIFICATION

SCALE SYSTEMS, PREPACKAGED FOOD

This specification is approved by the US Army Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers automatic electrically operated scale systems for weighing, computing prices and issuing labels for application to prepackaged foods and digital computing scale with labeling printer. The type I system shall issue labels automatically and when equipped with a conveyor table shall apply the labels automatically to the prepackaged foods.

1.2 Classification. The scale systems covered by this document shall be of the following types, classes and styles as specified (see 6.2).

Type I - Automatic scale system with memory and printer
  Class 1 - for use without universal pricing code
    Style A - with conveyor and index labeler
    Style C - with open table for manual operation (separate from wrapping machine)
  Class 2 - for use with universal pricing code
    Style B - with conveyor and index labeler
    Style D - with open table for manual operating (separate from wrapping machine)

Type III - Dual display digital computing scale with printer
  Class 1 - for use without universal pricing code
  Class 2 - for use with universal pricing code

Type IV - Checkstand scale

Type V - Produce scale

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6670

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-665	-	Boxes: Paperboard, Metal Edged and Components
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-60	-	Tape: Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-W-43644	-	Wrapping, Weighing and Labeling Units, Meat

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

OTHER GOVERNMENT DOCUMENTS

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook 44 - Specifications, Tolerances, and Other Technical Requirements for Weighing and Measuring Devices

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

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UNDERWRITERS LABORATORIES, INC.

Electrical Appliance and Utilization Equipment List

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Rd., Northbrook, IL 60062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The scale system shall be the manufacturer's commercially current standard product except for any changes necessary to comply with document requirements.

3.2 Safety standards. The scale systems shall conform to the applicable standards of the Underwriters Laboratories, Inc.

3.2.1 Certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the scale systems furnished under this document conforms to the applicable UL Standards. Acceptable evidence of meeting the applicable requirements of these standards shall be a listing in the UL, Electrical Appliance and Utilization Equipment List or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the scale systems have been tested and conform to applicable UL Standards.

3.3 Materials and components. Materials and components shall conform to the applicable documents and requirements specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all provisions of the document.

3.4 Design and construction.

3.4.1 Type I, class 1 scale system. The scale system shall consist of a digital readout scale with platform, computing mechanism, printer and rack, automatic electrically operated scale system, for weighing, computing prices and issuing labels from the printer. The scale shall be constructed so as to provide a digital

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readout of price, total weight, and price per pound on a display system. Scales shall include solid state integrated circuitry and solid state components. Range shall be from 0 to 30 pounds minimum, the computing of the value based on the price per pound from $0.01 through $99.99 in 1 cent graduations with display on operator side. Storage rack for commodity library shall be provided atop the scale or printer, with minimum capacity of 2,000 names to include all basic retail meat identity standard cuts or produce standard items. Memory file shall be capable of storing complete information on 500 commodities from one library. Information will include commodity name, USDA grade, price, tare, date, and shelf life. Printer shall be capable of printing information contained in file for any commodity stored in file and shall also print open pull date and machine operator code. The system shall be capable of providing total package count, net weight, and total dollar amount at the end of each run or on a daily basis total as required. The automatic labeling mechanism shall be designed to work with automatic pre-pack scale system for indexing, weighing and labeling of trayed packages. Corrosion-resistant steel, porcelain enameled carbon steel, epoxy coated or anodized aluminum shall be used for the outer covers. The scale system shall be capable of accurately weighing a load in accordance with test procedure and tolrances of NBS Handbook 44 when tested as specified in 4.3.1. The scale shall have a means to adjust zero in accordance with NBS Handbook 44.

3.4.1.1 Type I, class 2, scale system. The type I, class 2 scale system using the universal code shall consist of the following: A modular component scale system with a counter type platform scale, solid state controlling mechanism, with electronic memory and printer with universal bar code printing capability. The controlling mechanism shall incorporate a back-up battery for memory retention up to 20 hours in the event of an external power failure. The components operate as a system to automatically weigh, compute, print, and issue labels, and collect total data for management use. The platform scale shall provide for a digital readout of unit price, total weight in pounds, and total price in dollars. The system shall be electronically programmable with a minimum memory capability of 500 commodity description with price lookups. The size of the platform shall be essentially the same as the full scale base. The system shall weigh in increments from zero to 30 pounds. Price per pound ranges from $0.01 through $99.99. Maximum computed value per package shall be $999.99. Light emitting diodes shall display a minimum of five digits. The printer shall automatically deliver printed, heat seal or pressure-sensitive labels displaying the date, with minimum of 26 characters of description, total weight, unit price, total dollars, and universal pricing code bar code symbols. Provisions shall be made for tare weight. Maximum dimensions for the scale shall not exceed 20 inches in width, 20 inchs in depth and 19 inches in height. Maximum dimensions for the printer shall not exceed 19 inches in width, 16 inches in depth and 16 inches in height. Power requirements are 120 Volts, 60 Hertz, single phase, unless otherwise specified (see 6.2). Scale systems shall comply with the National Bureau of Standards Handbook 44.

3.4.1.1.1 Type I, scale. The type I, scale shall have means for entering and registering tare weight from zero to 8 ounces minimum. Leveling means shall be provided.

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3.4.1.1.2 Type I, scale platform. The type I, scale platform shall be rectangular with a surface area of not less than 175 square inches and a side of not less than 11 inches.

3.4.1.2 Type I, class 1 and 2, label. The type I, class 1 and 2 label printer shall be capable of producing and activating a label automatically by thermostatically controlled heating means for application to the prepackaged item. The labeler shall have a counter for recording the number of labels provided. The information contained on the label shall be such as weight, price per pound, computed value, store name product identification, grade, date code and operator or department code. The label detector shall be equipped with an alarm and an automatic shut down device to alert the operator when an interference occurs with the issuance of a label to the product. The powered roller package indexes shall feed package onto the weighing platter requiring no pusher arm contact with the package until weighing is complete, thus eliminating the possibility of incorrect weights.

3.4.1.3 Type I, class 1, style A, scale system. The type I, class 1, style A, scale system in addition to the components specified in 3.4.1 shall have a powered roller or conveyor and index labeler for automatic application of the label to the prepackaged item with means for a simple change over to manual label application for a large odd shaped items. The index labeler shall be designed to work with the scale system for indexing, weighing and labeling of prepackaged foods. The power roller or conveyor package shall feed the package onto the weighing platter and requiring no pusher arm contact with the package until weighing is completed. The scale system shall be designed to accommodate the wrapping machine covered by MIL-W-43644 (system III) as specified (see 6.2).

3.4.1.4 Type I, class 1, style C, scale system. The type I, class 1, style C, scale system shall consist of the components specified in 3.4.1 and shall produce a label for manual application to the prepackaged item. The manual operation, independent of the wrapping machine, two open style tables shall be furnished to support the scale and printer at the normal working height.

3.4.1.5 Type I, class 2, style B, scale system. The type I, class 2, style B, scale system shall be as specified in 3.4.1.3 except that the index labeler is for use with type I, class 2, scale system.

3.4.1.6 Type I, class 2, style D, scale system. The type I, class 2, style D, scale system shall be as specified in 3.4.1.4 except that the index labeler is for use with type I, class2, scale system.

3.4.2 Type III, classl, scale system. The type III, class 1, scale system shall consist of a scale with platform and a labeler. The design shall provide for the weighing of an item in a range of 0 to 30 pounds minimum, the computing of the value based on the price per pound with a range from 1 cent to 99 dollars and 99 cents per pound inclusive, in 1 cent graduations and the presenting of positive information digitally on both customers and operators side. The scale system shall be capable of accurately weighing a load in accordance with test procedure and tolerances of NBS Handbook 44 when tested as specified in 4.3.1. The scale shall have a means to adjust to zero in accordance with NBS Handbook 44.

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3.4.2.1 Type III scale. The scale shall be constructed so as to provide a digital readout of price, total weight, and price per pound on a dual display system. Scales shall include solid state integrated circuitry and solid state components. The scale shall provide automatic tare, within the scale capacity, when the operator activates the tare control. The scale shall not exceed 95 pounds in weight.

3.4.2.1.1 Type III scale platform. The platform shall be made of corrosion-resistant steel, epoxy coated or anodized aluminum and shall be rectangular with a surface area of not less than 135 square inches and a side length of not less than 11 inches.

3.4.2.2 Type III labeler. The labeler shall print and issue a label on pressure-sensitive stock tape with the price per pound, weight in pounds and the total price of the item contained thereon. The printed label shall be obtained by activating the print control. The labeler shall not exceed 9 inches width, 19 inches in depth and 13 inches in height or 30 pounds in weight.

3.4.3 Type III, class 2, system. The type III, class 2, scale system shall be as specified in 3.4.2 except that the universal pricing codes are to be furnished with the scale and printer.

3.4.4 Type IV, checkstand scale. The type IV, checkstand scale shall have a capacity of 30 pounds in graduations of 0.01 pounds with a stainless steel platter. The scale shall provide weight data to an electronic cash register on demand and the weight is multiplied by the price per pound from the register. The scale shall be equipped with a post mounted digital light emitting diodes display module approximately 12 inches above the scale platter for customer viewing convenience. The display module is remotely connected to the scale by interconnecting plug in cable and mounting bracket suitable for any horizontal surface. Features such as automatic zero maintenance, zero adjust push button, scale motion detertion, digital weight filter, display flashing for zero push button actuation, metric capability, and platter support system to assure accurate weights from all the paints of the weight platter shall be built in the scale. Accuracy shall meet the National Bureau of Standards Handbook 44. The maximum overall dimensions shall be 17 inches in width, 17 inches in depth and 19 inches in height.

3.4.5 Type V, produce scale. The type V, produce scale shall have a capacity of 30 pounds in graduations of 0.01 pounds with a stainless steel platter. The vendor and customer displays are to be flexible for convenient postioning. The display shall be a light emitting diode with an instant price caluation. The scale shall provide for pound entry, instant total price calculation for 4 digit pricing ($0.01 to $99.99 per pound), 5 digit total pricing to $999.99, tare weight. The scale shall comply with the National Bureau of Standards Handbook 44.

3.5 Electrical characteristics. When specified (see 6.2), type I and III scale systems shall operate on a nominal 120-volt, 60-Hertz, single phase system. The power supply cord shall be a 3-wire cord, 6 foot minimum length, with a 3-pronged plug. Unless otherwise specified (see 6.2) for overseas requirements, the scales shall operate on a nominal 220 Volts, 50 Hertz, 1 Phase electrical supply.

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3.6 Performance.

3.6.1 Overload. Type I and III scale system scales shall weigh a load equal to 125 percent of the capacity of the scale with no visible deformation, breakage, misalignment and failure to return to its intial zero position, when tested as specified in 4.3.2.

3.6.2 Operational. Type I and III scale systems, when tested as specified in 4.3.3:

(1) Shall operate smoothly with no binding, malfunction, visible bend or deformation.
(2) Shall produce accurate, clear and legible lables with distinct printing, containing specified information (see 3.4.1.1, 3.4.1.2 and 3.4.2.2).
(3) Shall register accurate tare weight settings (see 3.4.1.1.1 and 3.4.2.1).

3.7 Finish. The exterior surfaces of the scale system unless otherwise specified herein shall be finished in accordance with the manufacturer's commercial practice. The interior surfaces and working parts shall be protected against corrosion in accordance with the manufacturer's commercial practice.

3.8 Marking. The scale systems shall be marked by a decalcomania (decal) or permanently affixed metal tab that shall include the manufacturer's name or trademark, model number and electrical characteristics.

3.9 Lubrication. The contractor shall affix to each conveyor a lubricating chart (metal or decal) indicating all parts requiring lubrication, grade of lubricant required and time interval when lubrication shall be accomplished.

3.10 Workmanship. The scale system shall be free from fractures, splits, fractures, tears, dents, creases, bows, miscasts, deteriorations or malformations. There shall be no sharp edges, burrs or slivers. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray,, sag, imprint or embedded foreign matter. All bolts, studs, nuts and screws shall be drawn tight.

3.10.1 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, continuous, smooth and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds.

3.10.2 Electric wiring. Wiring shall be cut, abraded or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 Initial production item testing. The initial production item shall be tested as specified in 4.3.1, 4.3.2, and 4.3.3. Any failure of the initial product item shall be cause for rejection of the item, cause for necessary design and construction changes, and cause for repetition of the above testing on the next produced item. The testing process shall be repeated until an item successfully passes all the tests.

4.2.3 End item inspection.

4.2.3.1 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of one end item. The sample unit shall be one end item. The inspection level shall be level II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified		X
	Blistered, peeled, chipped, or area of no film		X
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign material imbedded		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated, or malformed	X	
	Component misplaced or not in proper alignment		X
	Component loose		X
	Sharp burr or edge or sliver		X
Type I	Tare weight range not from 0 to 8 ounces	X	
	Levelling means missing	X	
	Platform area less than 175 square inches	X	
	Platform side length less than 11 inches	X	
	Weight range not from 0 to 30 pounds	X	
	Price range per lb not $0.01 per lb to $99.99	X	
	Means for entering and allowing for tare weights missing	X	
Labeler	Counter for recording number of labels missing	X	
Type I, class 1, style A	Means for automatic application of label on prepackaged item missing	X	
Type III	Weight range not 0 to 30 pounds	X	
	Price per lb not $0.01 per pound through $99.99 per lb	X	
	Maximum computed value per packaged not $999.99	X	
	Platform area less than 135 square inches	X	
	Platform side length less than 1 inch	X	
	Incomplete or incorrect digital readout data	X	
Type III labeler	Size not as specified	X	
	Weight not as specified	X	
	Incomplete or incorrect label data	X	
Electrical	Power cord not a three-wire cord with a 3-pronged plug and minimum 6-foot length	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship, general (applicable to all components and assemblies) (cont'd)			
Marking	Marking with manufacturer's name or trademark, model number and electrical characteristics missing	X	
Lubrication	Lubrication chart for conveyor missing	X	
Welding	Welds not sound, continuous, smooth or free from porosity, cracks, incomplete fusion and deformation	X	
Bolts, studs, nuts and screws and other types of threaded fasteners	Scale or flux deposits not removed		X
	Stripped, broken or missing	X	
	Loose		X
Electric wiring	Wiring cut, abraded or not properly installed or joined	X	
	Adequate slack not provided for wiring to relieve strain		X

4.2.3.2 End item dimensional examination. The end item shall examined for conformance to the dimensions specified in this document. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of end item. The sample unit shall be one end item. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.3.3 End item testing. Each production end item shall be tested for operation as specified in 4.3.3. Failure to pass any test shall be cause for rejection of the end item.

4.2.3 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping

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container fully packed except that it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Examination for closure and strapping shall be in accordance with the appendix of the applicable container specification.

TABLE II. Packaging defects

Examine	Defect
Markings (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application
Preservative application	Improperly applied or missing
Materials	Component missing, damaged, or otherwise defective
Contents	Number per container is not as specified

4.2.4 Code and standard compliance. Certification that all the end items meets the requirements of 3.2.1 shall be furnished and will be acceptable to the Government representative.

4.3 Methods of inspection.

4.3.1 Shift test. An accurately determined known test weight (see 4.3.3) shall be centered successively at four points equidistant between the center and the right, left, back and front edges of the scale platform, weighed and checked for accuracy to determine compliance with the requirements of 3.4.1 and 3.4.2. Any nonconformance with the requirements shall constitute failure to pass the test.

4.3.2 Overload test. A weight equal to 125 percent of the scale capacity shall be placed on the scale platform. The weight shall be removed and the scale shall then be examined to determine compliance with the requirements of 3.6.1. Any nonconformance with the requirement shall constitute failure to pass this test.

4.3.3 Operation test. Type I and type III scale systems shall each be tested by weighing accurately determined known test weights of approximately equal weight. The system scale shall be loaded with the test weights one at a time, accumulatively, until the maximum capacity of the scale is reached. Each load shall be weighed; the value computed respectively, at prices per pound of 50 cents, 1 dollar, 1 dollar and 50 cents, 2 dollars, and 2 dollars and 50 cents; a label produced and the machine operation and accuracy of the label information checked to determine compliance with the requirements of 3.6.2 (1) and (2). The number of test weights shall be equal to the maximum capacity of the scale divided by 5. Tare weight settings shall be checked weighing a test weight with each of three different weights of packging material representing respectively a low, medium, and high tare weight in the scale range to determine compliance with 3.6.2 (3). Any nonconformance with the requirements shall constitute failure to pass this test.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly.

5.1.1.1.1 Label printers. Each label printer shall be disassembled to reduce cubage. Disassembly shall be limited so that no special skills or special tools are required for reassembly. Attaching hardware shall be placed in a cotton drawstring bag and the bag closed and secured to one of the mating parts.

5.1.1.1.2 Accessories. Accessories required to support the system shall be disassembled as specified in 5.1.1.1.1.

5.1.1.2 Preservation application.

5.1.1.2.1 Label printers and accessories. All exposed, uncoated ferrous metal surfaces, except corrosion-resisting steel, of each label printer, and supporting accessories shall be cleaned method C-1, thoroughly dried, and coated with type P-14 preservative in accordance with MIL-P-116. Surfaces from which preservative might be removed by contact with packaging materials shall be wrapped or covered with greaseproof barrier material conforming to type I or II, grade A of MIL-B-121. The greaseproof wrap or cover shall be secured with jute, sisal, manila, or cotton cord or twine or tape conforming to PPP-T-60 or PPP-T-45, or shall be heat-sealed with heat sealable barrier material is used.

5.1.1.2.2 Scale, labeler or label server. Movable components of each scale and each labeler or label server shall be secured in a fixed position in a manner as to prevent movement or damage during transit. Each scale and each labeler or each label server shall be individually preserved method II-b of MIL-P-116. Boxes used shall conform to type CF, variety optional, class domestic of PPP-B-636.

5.1.1.3 Unit packing.

5.1.1.3.1 Label printers. Each label printer, disassembled and preserved as specified in 5.1.1.1.1 and 5.1.1.2.1, respectively, shall be unit packed in a snug-fitting fiberboard box conforming to style RSC, type CF, variety optional, class domestic of PPP-B-636. Inserts made of the same material as the box, shall be used to immobilize the contents during transit. Closure shall be in accordance with the appendix of the box specification.

5.1.1.3.2 Accessories. Accessories, disassembled and preserved as specified in 5.1.1.1.2 and 5.1.1.2.1, respectively, shall be unit packed as specified in 5.1.1.3.1.

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5.1.1.3.3 Commodity card. Each commodity card shall be placed in a paper envelope.

5.1.1.3.4 Electrical cables. Electrical cables shall be compactly coiled and the coil secured with jute, sisal, manila, or cotton cord or twine.

5.1.2 Commercial. The scale system shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Scale systems. Each complete scale system, less the conveyor table for the type I, style A scale, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type of PPP-B-601, or class 2, style 2 or 4 of PPP-B-621. Contents of each shipping container shall be blocked, braced, and cushioned as required to prevent movement or damage during transit. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.1.2 Conveyor table. Each conveyor table shall be packed in a snug-fitting shipping container as specified in 5.2.1.1. Contents of each shipping container shall be blocked, braced, cushioned as required and the container provided with a case liner as specified in 5.2.1.1. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2 Level B packing.

5.2.2.1 Scale system. Each complete scale system, less the conveyor table for the type I, style A scale, preserved as specified in 5.1 shall be packed as specified in 5.2.1.1, except that shipping containers shall conform to domestic type, style A or B of PPP-B-601, or class I, style 2 or 4 of PPP-B-621, and the use of case liners shall not be required.

5.2.2.2 Conveyor table. Each conveyor table shall be packed as specified in 5.2.1.1, except that shipping containers shall conform to domestic type, style A or B of PPP-B-601, or class I, style 2 or 4 of PPP-B-621, and the use of case liners shall not be required.

5.2.3 Commercial packing. Scale systems, preserved as specified in 5.1, and conveyor tables shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Special handling marking shall apply.

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6. NOTES

6.1 Intended use. The scale systems are intended for use in military commissary stores to weigh, compute and issue printed labels for various prepacked food items.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Type, class and style scale system required (see 1.2).
(c) Wrapping machine required for type I, class 1, style A scale system (see 3.4.1.3).
(d) When other electrical characterics are required (see 3.5).
(e) Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any requirement for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.2.2 Equipment. The equipment to be acquired will be of the same type, capacity, anddesign that has been used in successful commercial supermarket operation for a minimum time of at least 2 years and is supported by a service organization providing service 24 hours a day, 7 days a week with response to requests for service within 4 hours during normal business hours.

6.2.3 Maintenance requirements. The vendor shall be capable of providing parts and maintenance of equipment within or adjacent to installation within 4 hours. Service of warranty must be available within same area to facilitate reasonable delivery and pickup for compliance of service warranty.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - GL

Review activity:

DLA - GS

Preparing activity:

Army - GL

Project No. 6670-A156

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-43147D(GL)	Scale Systems, Prepackaged Food

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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